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MILITARY STANDARD

TEST AND INSPECTION

OF

TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

FSC 3930

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ARMED FORCES SUPPLY SUPPORT CENTER
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TEST AND INSPECTION OF TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

MIL-STD-266A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 22 June 1961.

2. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Armed Forces Supply Support Center, Washington 25, D. C.

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TEST AND INSPECTION OF TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

1. SCOPE

1.1 SCOPE. This standard covers the testing and inspection procedures to be used as a guide in determining compliance with performance requirements of equipment listed in Specification MIL-T-3997.

2. REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this standard to the extent specified herein. When conflicts exist between this standard and any referenced document, the latter shall govern.

SPECIFICATIONS

MILITARY

MIL-T-3997 — Truck, Fixed Platform, 2,000 Pounds Capacity, Powered, Electric, Pneumatic Rubber Tires.

PUBLICATIONS

MILITARY

NAVSANDA Publication 284 — Storage and Materials Handling.

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS— NOT APPLICABLE

4. GENERAL REQUIREMENTS

4.1 REQUIREMENTS FOR APPROVAL. Items of equipment tested and inspected under this standard shall pass the following tests and inspections outlined herein:

A. Pre-Test
B. Performance Test
C. Post-Test

Examination and testing, whenever practicable, shall be generally conducted at the contractor's plant. If the contractor does not have approved test facilities, the performance tests and post-tests will be tested at a Government test facility. In any case, pre-inspection and pre-tests shall be conducted at the contractor's plant and a copy of the test results shall be forwarded by the Government Inspector with the truck to the designated testing facility. Final determination of the test site shall be left to the discretion of the contracting officer.

4.2 EQUIPMENT OPERATORS. When tests are conducted at a Government test facility, equipment operators shall be furnished by the Government to assist the test officer in conducting test and inspection.

4.2.1 Operator qualification. Government operators assigned for testing a type and model of equipment under the provisions of this standard shall have satisfactorily completed the applicable portions of the course entitled "On-the-Job Training Course for Operators of Materials Handling Equipment," NAVSANDA Publication 284. He shall have available, and show to contractor's representatives upon request, "Standard Operator's Permit" (DD Form 313), or its equivalent.

4.2.2 Operator **indoctrination.** Operators provided for testing any machine under provisions of this standard shall be required to drive that machine through the appropriate test course until he has achieved sufficient proficiency in operating the machine. This warmup period shall last for no more than 2 hours per driver. The contractor's representatives will be permitted to inform each

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driver of the proper technique to be used in operating the machine to achieve maximum performance.

4.3 **TEST LOCATION.** The test location will be designated by the contracting officer.

4.4 **TEST OFFICER.** The test officer shall be appointed by the Department of Defense and shall exercise full authority in directing the test.

4.5 **AMBIENT TEMPERATURE.** Temperature of the surrounding atmosphere may range from 40° F. to 104° F., inclusive. If the temperature drops below or rises above this range, the test may be discontinued at the discretion of the test officer and the manufacturer until the temperature returns to this range. The Government assumes no responsibility for adverse effects that may result from testing in other than the range of temperatures specified herein.

4.6 **CONTRACTOR'S REPRESENTATIVES.** Any contractor whose machine is being tested in accordance with the provisions of this standard will be permitted to have representatives witness the test and inspection and assist the test officer (upon request) with no obligation to the Government.

4.7 **OFFICIAL TEST REPORT.** An official report of the test and inspection will be prepared. It will include records of the pre-test, performance tests, post-tests and such other tests as were deemed necessary together with a statement from the test officer indicating whether the machine has passed or failed. The test officer will prepare as many copies as directed by the contracting officer.

5. DETAILED REQUIREMENTS

5.1 PRE-TEST. The pre-test shall consist of 3 principal parts, each of which is described below:

A. Maintenance, Service and Inspection
B. Inspection Check List
C. Inspection Test

5.1.1 **Maintenance, service and inspection.** Maintenance, service and inspection shall be conducted in accordance with Maintenance, Service and Inspection Sheet.

5.1.2 **Inspection check list.** All inspections and other procedures needed to mark the questions contained in the check list shall be performed in conjunction with, or immediately subsequent to, the initial inspection test referred to in 5.1.3. Each of the questions contained in the pre-test check list shall be answered by the test officer by placing an "X" in the appropriate column. After completion of the check list, any "X" in the "NO" column shall be cause for rejection of the unit under test.

5.1.3 **Inspection tests.** Inspection tests shall be conducted in accordance with procedures outlined in the test forms to determine compliance or non-compliance with contractual requirements.

5.1.4 **Standard procedures.** All portions of the pre-test shall be completed satisfactorily at the contractor's plant prior to the performance test. When performance testing is conducted at a Government test facility, pre-tests not conducted at the contractor's plant will be conducted by the Government upon request from the cognizant Government Inspector. Inspections and tests under the provisions of this standard requiring use of standard loads shall utilize the rated load for the required capacity. Loads may be of any desired material, but they shall be uniformly distributed over the load carrying platform of the unit under test. A tolerance of minus 1% plus 0% of the specified test loads will be permitted.

5.2 PERFORMANCE TEST

5.2.1 **Layout of course.** The test course shall be set up in accordance with the provisions of Drawing A. The length of each leg and number of curves in each leg may be varied, so long as the total length of the course and its general proportions are maintained. Aisle or road widths shall be 12 feet. The perimeter of the course shall be encompassed

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with suitable barriers or indicators. Rubber plyons, or other suitable markings shall be placed at all corners and other critical points to assist in keeping the machine under test within the course limits.

5.2.1.1 **Ramp inclusion.** The course should be laid out to include natural grades of the slope and approximate length specified. Only if this is impossible to do shall a ramp be constructed. When constructed, it shall be in accordance with Drawing C.

5.2.1.2 **Obstacle inclusion.** Obstacles shall be included in the course layout as indicated on Drawing A. Iron, wood, cement or steel blocks may be used. A guideline 15 feet long shall be provided to the left of the centerline of the course and parallel to it. This line shall be at a distance from the course centerline equal to one-half the distance between front wheel centers when two steer wheels are used. This spacing and arrangement will allow the left wheels of the machine under test to pass over the first pair of blocks and then the right wheels to pass over the second pair of blocks while the machine is moving forward in a straight line with the left front wheel tracking on the guideline.

5.2.1.3 **Course surface.** The test course shall be covered with hard packed earth or gravel, with a surface permitting tire impression ¼ inch deep when the carrier is transporting a capacity load, or any other surface or paving which meets this requirement. The course shall be free of snow, ice, any non-planned obstacles or foreign materials during the course of test.

5.2.1.4 **Minimum lighting requirements.** Minimum general illumination of the course shall be 5-foot-candles on the operating level. However, lights of the truck shall be on at all times during the test. The light switch shall be turned "off" and "on" at the beginning of each lap.

5.2.2 **Changing batteries.** Each machine under test shall utilize not more than two sets of fully charged batteries during the course of each operating 8-hour day of the performance test. However, the time required to change batteries under the provisions of this paragraph shall be excluded from the operating time. Batteries used shall have minimum ratings as specified in Specification MIL-T-3997.

5.2.3 **Speed of negotiating course.** Each machine undergoing test should traverse each part of the performance test course at the maximum safe practical speed.

5.2.3.1 **Laps per hour and per day.** Each truck shall complete 3000 complete circuits of the test course. Circuits (laps) shall be conducted on a continuous basis during each 8-hour day with 8 circuits being equivalent to one hour of operation. Not more than two sets of fully charged batteries of the capacities specified shall be utilized.

5.2.4 **Duration of the performance test.** The performance test specified in this standard shall be conducted on a continuous basis each operating 8-hour day until the number of laps required are completed.

5.2.5 **Periodic maintenance.** On the fourth operating day of the performance test the "Maintenance, Service and Inspection" shall be given according to the provisions of "Maintenance, Service and Inspection Sheet". This procedure shall also be carried out after the machine under test has completed performance test requirements. However, no repairs or replacements of parts shall be permitted.

5.2.6 **Method of negotiating course.** The machine under test shall begin each lap at the point labeled "START" on Drawing A. It shall have been loaded with rated load prior to start of the test. As the operator starts the machine he shall operate the horn for approximately 1 second; turn on headlights, turn off and turn on. Proceed in a forward direction along the portion of the course marked "A" until it reaches the first turn area marked "B" completing a 360° turn in a clockwise direction and continue to stopping point. After stopping, the machine shall make a 90° turn and proceed onto the natural slope or ramp. The machine under test shall pro-

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ceed up the slope or ramp to the position labeled "Stopping Point Under Load" on the drawing. Here the driver shall stop the machine and then resume mounting the slope or ramp. After reaching the top of the slope or ramp, the machine shall proceed along the main aisle in the portion of the course marked "E" and "F" until it reaches the portion of the course marked "G" which contains the obstacle test.

The obstacle test shall be traversed by the machine being driven through this area in a straight line, with its left front wheel (when two steer wheels are used) on the guide line, so that its left wheels pass over the first pair of blocks and its right wheels pass over the second pair of blocks. For truck incorporating a single steer wheel, one pair of blocks shall be placed such that they will constitute an obstacle for the single steer wheel. This part of the test may be conducted at low speed. However, the machine shall not be stopped prior to going over the blocks. After the obstacle test has been traversed the machine shall proceed along the main aisle in the portion of the course marked "H" until it reaches the final turn and stop.

Here the machine shall make a tight 360° turn, in a counterclockwise direction at maximum acceleration for the first 90° of the turn, and then stop upon completion of the turn.

The driver shall continue to maneuver the machine around the course, in the manner described above, in accordance with 5.2.3.

On alternate days, the course shall be traversed in the opposite (or clockwise) direction. On these days the test procedures shall be reversed.

The "Hourly Time Record Sheet" (BUSANDA Form 2522) shall be filled out for each operating hour of the performance test.

5.2.7 Operating limitations. The truck under test shall not be permitted to leave the bounds circumscribed in Drawing A.

5.2.8 Failure during performance testing. During the endurance testing, minor adjustments or repairs will be permitted at the discretion of the test officer provided they are not repetitive in nature. Minor breakdowns which are repetitive and major breakdowns shall be cause for rejection of the truck. A major breakdown is defined as a failure in a major component of the truck or one requiring more than one hour to repair using conventional shop tools and practices.

5.3 **POST-TEST.** The post-test shall be conducted after a machine has completed the performance test or, if it fails the performance test, the inspection portion of this procedure shall be conducted after such a failure. The post-test consists of two parts:

A. Post-Test Inspection Tests.
B. Post-Test Inspection

5.3.1 **Post-Test inspection tests.** These tests shall be performed in accordance with the applicable procedures outlined in the Test Forms. The truck should be given the maintenance, service and inspection in accordance with the applicable Maintenance, Service Inspection sheet. Each of the questions contained in the Post-Test Inspection Test Check Sheet shall be answered by placing an "X" in the column headed "Yes" or "No" as appropriate. After completion of the Post-Test Inspection Test Check Sheet any "X" in the "No" column will be cause for rejection of the truck under test.

5.3.2 **Post-Test inspection form.** The test officer shall examine the truck and make applicable notation on the Post-Test Inspection Form. Defects discovered during post-test inspection shall be corrected as recommended by the inspection officer.

5.3.3 **Commutator damage.** During the course of the post-test inspection maintenance personnel shall examine the commutator of the main drive motor(s). If any excessive pitting is found in this part, the test officer shall recommend rejection of the machine under test. A report of the findings of this examination shall be prepared by the test officer and shall become a portion of the official test report of the machine under test.

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PRE-TEST INSPECTION CHECK LIST

TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Description of Truck_____

Truck Mfr._____ Date_____

Model No._____ Registration No._____

Serial No._____ Recorded by_____

1. Are all certificates of compliance furnished? YES_____ NO_____

2. Does the truck comply with the applicable safety requirements? (Refer to Underwriters' Laboratories Publication, Standard for Safety, Power-Operated Industrial Trucks) YES_____ NO_____

   NOTE: Where there is a conflict between Underwriters' Standard for safety and and the specification requirements referenced in the contract, the specification shall govern.

3. Do the following truck components comply with all requirements of the specifications referenced in the contract?

	Yes	No	Remarks
A. — Electrical Equipment			
Wiring			
Resistors			
Insulation			
Enclosures and Shields			
Overcurrent Device			
Resistor Housings			
Thermal relays (Type EE trucks)			
Static Discharge Device			
Control Panel			
Conductors			
Traction Motor			
Thermal relays (Type EE trucks)			
Insulation			
Speed Controller			
B. — Battery and Battery Accessories			
Battery (When furnished)			
Volts			
Amperes			
Battery leads			
Battery connectors			

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	Yes	No	Remarks
B. — Batteries and Battery Accessories—*Continued*			
Battery compartment			
Length			
Width			
Height			
C. — Drive Unit			
Drive gears			
Shafts			
Axles			
Lubrication			
D. — Structure			
Chassis and frame			
Frame members			
Bracings			
Joints			
Bumper plate			
Deck			
Wheels			
Drive wheels			
Steer wheel(s)			
Tires			
Tubes (when used)			
Steering			
Rotation of hand wheel			
Adjustment			
Steer axle			
Wheel mounting			
Spring suspension			
Service brakes			
Adjustment			
Wheel cylinders			
Brake lining			
Master cylinder and pedal linkages			
Access hole opening			
Relay switch			
Parking brakes			
Seat			
E. — Controls and Instrumentation			
Protection against weather			
Position markings			
Distance between knobs			
Light switch			
Brake pedal			
Accelerator			
Directional controls			
Instruments			
Hour meter (note reading in remarks)			
F. — Accessories			
Headlight			
Focus			
Stoplight			
Operation			
Protection against damage			
Horn			
Fire extinguisher (when required)			

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	Yes	No	Remarks
G. — Bearings			
H. — Lubrication			
Reservoirs			
Accessibility			
Grease fittings			
I. — Treatment and Painting			
J. — Radio Interference Suppression			
K. — Nameplate			
Registration data			
L. — Repair and Maintenance			
Adjustment, servicing and replacement of parts			
Access openings			
M. — Technical Manuals			
N. — Workmanship			
Finish and appearance			
Rivet connection			
Welds			
Bolt connections			
Castings			

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TEST FORM No. 1

STEERING WHEEL
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether the force required to turn the steering wheel meets specification.

Test Description: Place the unloaded truck on a level concrete surface. Using a tension dynamometer, or equivalent, measure the force required to move the steering wheel.

Test Procedure:

1. Place truck on dry, level, smooth, trowelled or brushed concrete with steer wheel(s) in a straight-ahead position.
2. Assemble an adapter pulley having a root diameter equal to the steering wheel diameter and equipped with means for attachment to the wheel spokes.
3. Attach a tension dynamometer to the adapter pulley.
4. Grasp the dynamometer lead and apply a steady force, in a counter-clockwise direction and in the wheel plane, until the wheel turns.
5. Observe the maximum gage reading.
6. Perform 4 and 5 in a clockwise direction.

After completing test requirements above, conduct the following:

7. Raise front of truck and block in raised position.
8. Lock steerable wheel(s) in straight-ahead positions.
9. Apply through the tension dynamometer, in a clockwise direction, maximum specified force.
10. Repeat in counterclockwise direction.
11. Perform operations 9 and 10 four times.
12. Repeat operations 9, 10 and 11 with other steerable wheel(s) locked in straight-ahead position.
13. Complete data form.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

Question	Counter-clockwise Direction	Clockwise Direction
Maximum gage reading first half of inner steer angle		
Maximum gage reading at extreme ends of Steer		

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TEST FORM No. 1—(Continued)

Question	Yes	No
Does force required to turn the hand steering wheel comply with specified requirements?		
Is there no evidence of permanent deformation to steering wheel and related steering mechanism when applying the specified force with steerable wheel(s) blocked?		

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TEST FORM No. 2

WEAR OF STEERING MECHANISM
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine cumulative wear in the steering mechanism of the truck due to carrying out requirements of the performance test.

Test Description: With the truck stationary and without load, the amount of free play in the steering wheel is measured.

Test Procedure:

1. Place unloaded truck on a dry, level, concrete surface with its front wheel(s) parallel to the longitudinal axis of the truck.
2. Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semi-circular in shape and which has a radius of at least one inch greater than the radius of the hand steering wheel.
3. Mark a reference point on the upper portion of the outer edge of the hand steering wheel; mark a corresponding point on the fiber board templet.
4. Turn the steering wheel counterclockwise until the front wheel(s) of the truck begins to turn. Mark a point in the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.
5. Turn the steering wheel clockwise until the front wheel(s) of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.
6. Remove fiberboard templet from truck.
7. Place fiberboard templet on a flat surface and measure the distance between the two points made in accordance with operations 4 and 5 above. This distance, in inches, is the free play of the steering mechanism.
8. Complete data form.

DATA FORM:

Description of Truck _____

Truck Mfr. _____ Date _____

Model _____ Test Location _____

Serial No. _____ Recorded by _____

Free Play of Steering Wheel:

Before Performance Test _____ inches

After Performance Test _____ inches

Question	Yes	No
Is the free play of the steering wheel before the Performance Test not more than 3 inches?		
Is the free play of the steering wheel after the Performance Test not more than 1.5 times the free play of the steering wheel before the performance test?		

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TEST FORM No. 3

SPEED
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether truck meets the speed requirements of MIL–T–3997.

Test Description: Truck, with rated load, is accelerated through a 50 foot or more approach, timed through an 88 foot test length, and then brought to a stop. (This 88 foot length is used to simplify calculations as 88 feet/minute is 1 m.p.h.). Three runs are made forward in high speed. The data is recorded and average speeds determined.

Course Layout:

A————50′ or more————B
B————88′————C

Test Procedure:

1. Locate truck, carrying rated load, for forward run with front wheels at Point A.
2. Start forward test run and have truck at top speed by the time it reaches Point B.
3. Using fractional second timing device, determine time in seconds using in travelling from line B to line C and record on data form.
4. Perform operations 3 and 4 a second and third time.
5. Complete data form.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

Time in seconds, No. 1 test	
Time in seconds, No. 2 test	
Time in seconds, No. 3 test	
Total time	
Average time (⅓ of total time)	
Average speed in m.p.h. $\frac{60}{\text{average time}}$	

Question	Yes	No
Does average forward speed with load meet specified requirements?		

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TEST FORM No. 4

SLOPE ASCENSION — PARKING BRAKE TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether truck, when carrying rated load, can accelerate from a dead stop on a 15 percent grade and has sufficient underclearance to operate from a 15 percent slope to a horizontal surface. To determine whether the parking brake can hold truck with rated load on a 15 percent slope in both forward and reverse direction.

Test Description: Truck ascends a 15 percent grade in a forward direction on a dry concrete surface. Truck stops and starts up slope again. Horizontal approaches should be provided for the ramp.

Test Procedure:

1. Place rated load on platform.
2. Drive truck along horizontal approach to the 15 percent ramp.
3. Drive truck approximately half-way up the 15 percent ramp and bring to a complete stop.
4. Apply parking brake, release service brakes and observe whether truck remains stationary.
5. Release parking brake, and start truck from dead start and proceed up ramp and onto a horizontal surface.
6. Descend ramp. Stop on ramp and apply parking brakes, release service brakes, and observe whether truck remains stationary.
7. Proceed on to the horizontal surface.
8. Complete data form.

DATA FORM:

Description of Truck _____

Truck Mfr. _____ Date _____

Model _____ Test Location _____

Serial No. _____ Recorded by _____

Question	Yes	No
Can truck, when carrying rated load, stop on a 15 percent grade and start up again?		
Does truck have sufficient underclearance to permit operation up and down a 15 percent slope to a horizontal surface?		
Does the parking brake hold the loaded truck on the 15 percent grade in both forward and reverse directions?		

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TEST FORM No. 5

SERVICE BRAKE PRESSURE
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER-TIRES

Purpose: To determine whether brake pedal and brake system have the strength required by the specifications.

Test Description: Using a Bennett Feragen Corp. "Press-O-Meter", Model PM-35 or the equivalent, and suitable clamps, apply a pressure of 250 pounds to the brake pedal 10 times.

Test Procedure:

1. Attach the pressure gage to the brake pedal by a clamp to permit application of foot pressure to one side of the pressure gage.
2. Apply foot pressure to the gage until it registers 250 pounds and hold for 1 minute.
3. Repeat operation number two, 10 times.
4. Complete data form.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

Question	Yes	No
In 10 trials, can the brake system withstand a pedal thrust of 250 pounds without failure of any components?		

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TEST FORM No. 6

SERVICE BRAKE PRESSURE SENSITIVITY
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether a brake pedal pressure within the range of 75 to 125 pounds will develop a draw bar pull equal to approximately 18 percent of the gross weight of the truck (with rated load) while truck is being pulled.

Test Description: Pull truck with rated load over dry, smooth, trowelled or brushed concrete with parking brake released. Use a Bennett Feragen Corp. "Press - O - Meter" Model PM-35 or the equivalent, equipped with clamps. If the concrete or road surface does not have sufficient coefficient of friction, the surface may be brushed with Minnesota-Mining compound or other material to provide suitable coefficient of friction to prevent wheels from sliding while applying the required drawbarpull.

Test Procedure:

1. Record total weight of truck plus rated load.
2. Attach pressure gage to brake pedal, by means of clamps, in a manner which will allow the operator to apply foot pressure to face of the gage.
3. Attach a tension dynamometer to the front of the truck.
4. Attach one end of a block and tackle, or similar device for securing mechanical advantage, to the tension dynamometer, and the other end to an anchoring device.
5. Apply sufficient brake pedal pressure to prevent wheel from turning.
6. Apply tension to the dynamometer until force is approximate equal to 18 percent of the total weight of truck plus rated load.
7. Gradually reduce brake pedal pressure until wheels just begin to turn while maintaining the dynamometer tension. This pedal pressure shall not be less than 75 pounds nor more than 125 pounds.
8. Record brake pedal pressure and dynamometer readings.
9. Complete data form.

DATA FORM:

Description of Truck_____ Weight of Truck_____

Truck Mfr._____ Weight of Load_____

Model_____ Date_____

Serial No._____ Test Location_____

Recorded by_____

Brake Pedal Pressure	
Tension Dynamometer Reading	
18 Percent of Truck with Load	

Question	Forward		Reverse	
	Yes	No	Yes	No
Was a drawbar pull of 18 percent of weight of truck with rated load developed by a brake pedal pressure between 75 to 125 pounds?				

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TEST FORM No. 7

TURNING RADIUS
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether truck can turn in 90 degree intersecting aisle of specified width.

Test Description: 90 degree intersecting aisles of specified width are laid out. The truck must make a right hand turn within the specified aisle widths in one motion, without backing, and with no more than three trials. The test is repeated with the truck making a left hand turn.

Test Procedure:

1. Layout on dry, smooth, trowelled or brushed concrete two 90 degree intersecting aisles of specified width.
2. Drive truck at slow speed, in one motion and without backing, around the intersecting aisles to the right. Truck must negotiate the turn within the specified aisle width in not more than three trials.
3. Perform operation 2 with truck turning to the left.
4. Complete data form.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

Question	Yes	No
Was truck able to make a right hand 90 degree turn in intersecting aisles of specified width in one motion and without backing, in no more than three trials?		
Was truck able to make left hand 90 degree turn in intersecting aisles of specified width in one motion and without backing, in no more than three trials?		

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TEST FORM No. 8

ACCELERATION
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine acceleration of truck with and without load, on a level surface from a standing start.

Test Description: Truck is accelerated at maximum rate from a standing start, timed through a distance of 22 feet, and then brought to a stop.

Course Layout:

A————22'————B

Test Procedures:

1. Locate truck without load, with front wheels on Point A.
2. Station an observer at Point B. The observer shall be provided with a fractional second timing device.
3. The test leader shall be stationed at Point A.
4. Test leader shall start test by calling "mark". When this signal is given the truck operator starts the truck and the observer starts his watch.
5. The truck operator will maintain maximum truck acceleration through the entire 22 foot test run.
6. As the front wheels of the truck pass over Point B, the observer stops the timing device.
7. Record stop watch times on data form.
8. Perform operations 1 through 7 a second and third time.
9. Perform operations 1 through 8 with truck carrying rated load.
10. Complete data form.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

	Truck Empty	Truck Loaded
Time in seconds for 22 feet No. 1 test		
Time in seconds for 22 feet No. 2 test		
Time in seconds for 22 feet No. 3 test		
Total time		
Average time (⅓ of total time)		
Average time x average time (t^2)		
Acceleration A ($44 \div t^2$)		

Question	Yes	No
Does average acceleration meet specified requirements? Without load? With load?		

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TEST FORM No. 9

OVERLOAD TEST
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether the truck structure can withstand specified overload without permanent visible deflection or deformation.

Test Description: With truck on a level surface, specified overload shall be placed on the truck platform. There shall be no permanent deformation or failures of related structures.

Test Procedure:

1. Place truck on level surface.
2. With crane or forklift, the truck shall be uniformly loaded with specified overload. When placing the overload on trucks, personnel should stand clear and loads should be removed if there is evidence of immediate collapse or breakage in truck structure.
3. If truck accepts loads without immediate failure, let overloads stand for 10 minutes on truck.
4. Remove loads.
5. Inspect truck structure and frame for evidence of deformation, fractures, broken welds, etc.
6. Complete data form.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Registration No._____

Recorded by_____

Question	Yes	No
Is there no evidence of failure or deformation in truck structure?		

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TEST FORM No. 10

POWER TRAIN STATIC TORQUE TEST*
TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Purpose: To determine whether the drive line of the truck is designed to withstand 150 percent of the maximum stalled motor torque.

Test Description: With drive wheels locked, the required force is applied to a torque arm attached to a tension dynamometer. The torque arm shall be attached to the output shaft of the motor. This test may be performed as a bench test prior to assembly or with complete assembled truck at option of contractor.

Test Procedure:

1. Transmission shall be placed in lowest gear range.
2. Lock both drive wheels to prevent turning.
3. Clamp a torque arm of "L" feet to the output shaft of motor.
4. Obtain from manufacturer's published data, the stalled motor torque.
5. Apply to the torque arm, a force equal to $\frac{1.5T}{L}$.
6. Disassemble power train and examine for failures, fractures, or permanent distortion and deformation.
7. Complete data form.

*NOTE: This test is to be performed when specified either in the invitation for bid or contract.

DATA FORM:

Description of Truck_____

Truck Mfr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

Question	Yes	No
Is there no evidence of failure, fractures, permanent distortion or deformation?		

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POST-TEST INSPECTION TEST CHECK SHEET

TRUCK, FIXED PLATFORM, POWERED, ELECTRIC, PNEUMATIC RUBBER TIRES

Description of Truck _____

Truck Mfr. _____ Date _____

Model _____ Registration No. _____

Serial No. _____ Recorded by _____

Component	Applicable Test Form	Items to be Checked	Yes	No
Steering wheel	1	Does force to turn hand steering wheel comply with the specification?		
Wear of steering	2	Is the free play of steering wheel not more than 1.5 times the free play before the performance test?		
Speed	3	Does average forward speed comply with specified requirements?		
Slope ascension	4	Can trucks when carrying rated load, accelerate a 15 per cent slope from a dead stop?		
Parking brake	4	Can parking brake hold the truck with rated load on a 15 percent slope in forward and reverse direction?		
Service brake pressure	5	Can brake system withstand a pedal thrust of 250 pounds as tested without failure of any component?		
Service brake pressure sensitivity	6	Does pedal pressure of 75 to 125 pounds develop the specified drawbar pull?		
Acceleration	8	Can truck accomplish required acceleration?		
General		Any evidence of failure, excessive wear, or deformation in any component of the truck, or any defects which might cause failure during normal operation?		

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POST-TEST INSPECTION FORM

INSTRUCTIONS: Examine each item below in sufficient detail to determine any defects or excessive wear which might cause failure in normal operation. Place a check mark in column No. 1 or No. 2 as applicable. If column No. 2 is checked, also check column Nos. 3, 4 or 5. Additional remarks may be listed on a separate sheet of paper. Remarks should include details of adjustment, repair, or replacements indicated by check marks.

Description	(1) O.K.	(2) Not O.K.	(3) Adjust	(4) Repair	(5) Replace
1. Batteries (when furnished)					
Capacity					
Specific Gravity					
Blocking					
2. Brakes, Parking and Service					
Master cylinder					
Wear of brake lining					
Wheel cylinders					
Pedal linkages					
3. Contacts					
Condition of tips					
4. Drive Unit					
Differential					
Drive Axle					
Lubrication fittings					
Serviceability					
5. Electric Components					
Contactors					
Controller					
Accelerator control					
Interlocks					
Switches					
Battery connectors					
Motors					
Resistors & heat actuated devices					
Seat & brake switch					
Wiring					
6. Structure					
Chassis & frame					
Wheel mountings					
7. Steering assembly					
8. Springs					
9. Controls & instrumentation					
10. Accessories					
11. Workmanship					
12. Repair and ease of maintenance					

DATA FORM:

Truck Mfr._____ Date_____
Model_____ Test Location_____
Serial No._____ Registration No._____
Mechanic_____ Payroll No._____

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MAINTENANCE SERVICE AND INSPECTION SHEET MATERIALS HANDLING EQUIPMENT — ELECTRIC

The following inspections and checks shall be made prior to the performance testing and prior to the post testing. On the 4th and 8th days of the performance testing items 1 through 11 shall be checked.

1. Check all instruments for proper operation_____check free play in steering _____check horn operation.
2. Check brake pedal travel_____check fluid level_____adjust brakes if necessary_____check safety seat brake_____check linkage and operation _____inflate tire to proper pressure (cold) and remove all foreign matter from tires.
3. Inspect all lights.
4. Check battery_____ water level and specific gravity.
5. Inspect all safety equipment_____fire extinguishers, etc.
6. Check controller, contact tips, switches, operation.
7. Check motors brush wear, spring tension, commutator wear_____blow out carbon dust.
8. Check acceleration control_____speed_____general operation.
9. Inspect all wires and terminals; tighten and replace where necessary.
10. Lubricate chassis using manufacturer's reference charts_____replace missing or broken fittings where necessary_____make all necessary adjustments.
11. Inspect drive unit_____check for noise_____check oil level_____fill if necessary_____do not overfill.
12. Inspect steering gear_____check oil level_____fill if necessary_____do not overfill.
13. Check brake lining and drums, check wheel cylinders.

DATA FORM:

Description of Truck_____	
Truck Mfr._____	Date_____
Model_____	Test Location_____
Serial No._____	Type_____
Total hours_____	Recorded by_____

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HOURLY TIME RECORD
BSSARDA FORM 2522 (12-58)

TRUCK MANUFACTURER	MODEL	SERIAL NO.	DATE

TEST LOCATION	RECORDED BY:	CLOCK READINGS: (Start), (Finish)	REQUIRED LAPS/HOUR

LAPS NUMBER	WATCH READING	ELAPSED TIME	REMARKS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
TOTAL			AVERAGE LAP TIME_____

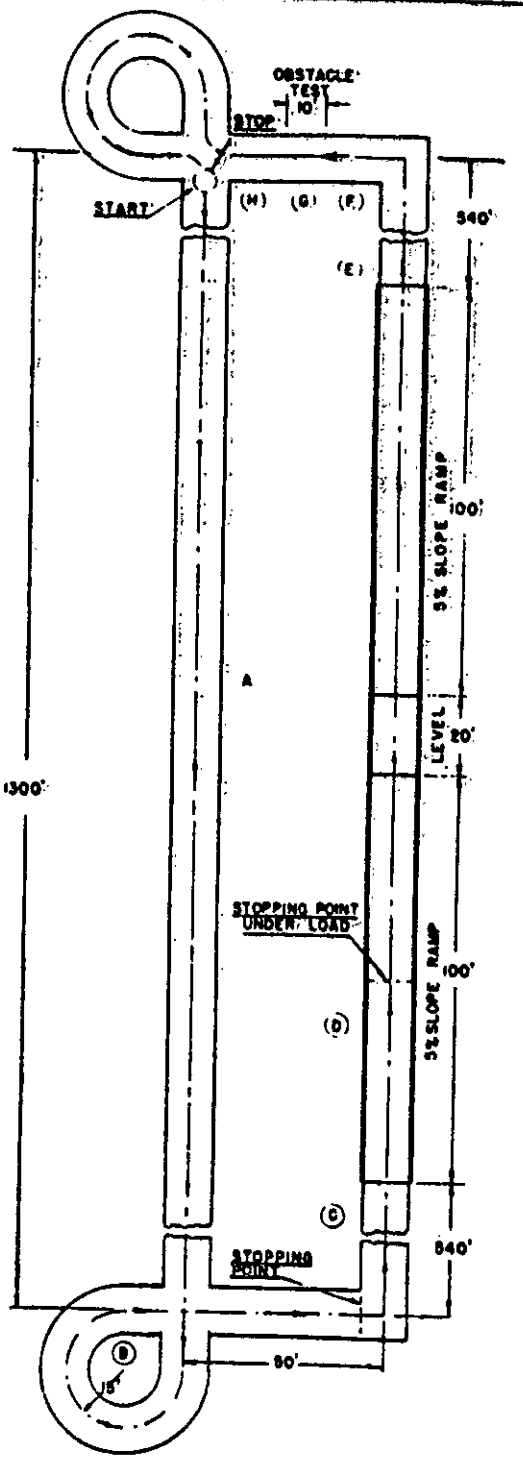
Instructions for Timing Performance Test

1. Start decimal minute stop watch at zero at beginning of each operating hour.
2. Record watch reading at end of each lap in "Watch Reading" space.
3. Subtract watch reading at end of preceding lap from reading at end of lap. Record differences in "Elapsed Time" space.
4. If equipment under test completes the required number of laps prior to completion of an operating hour, it shall be stopped until start of next hour, and then resume performance test.

NOTE: This form is for use in conjunction with Military Test and Inspection Standards during materiels handling equipment performance test.

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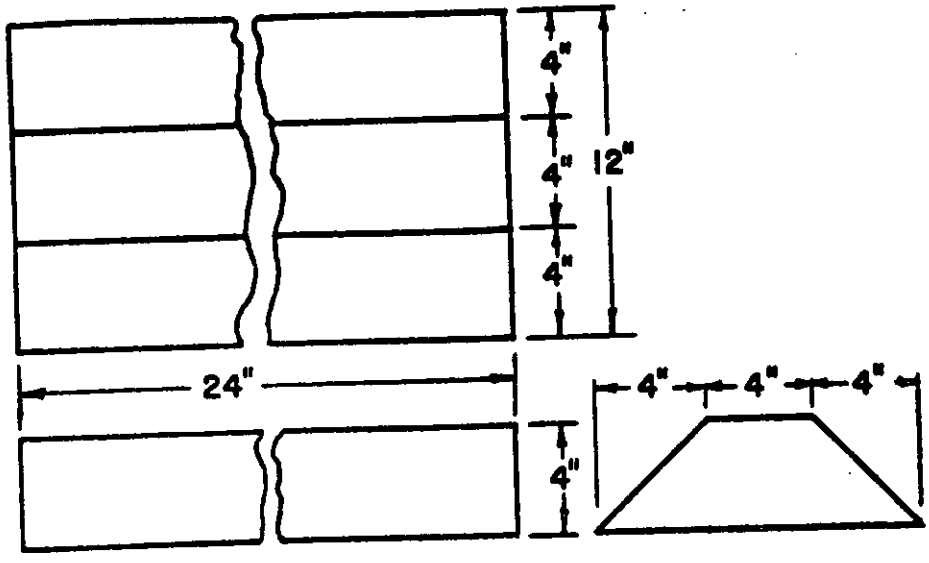
DRAWING "A"
OUTDOOR TEST COURSE FOR PLATFORM TRUCKS



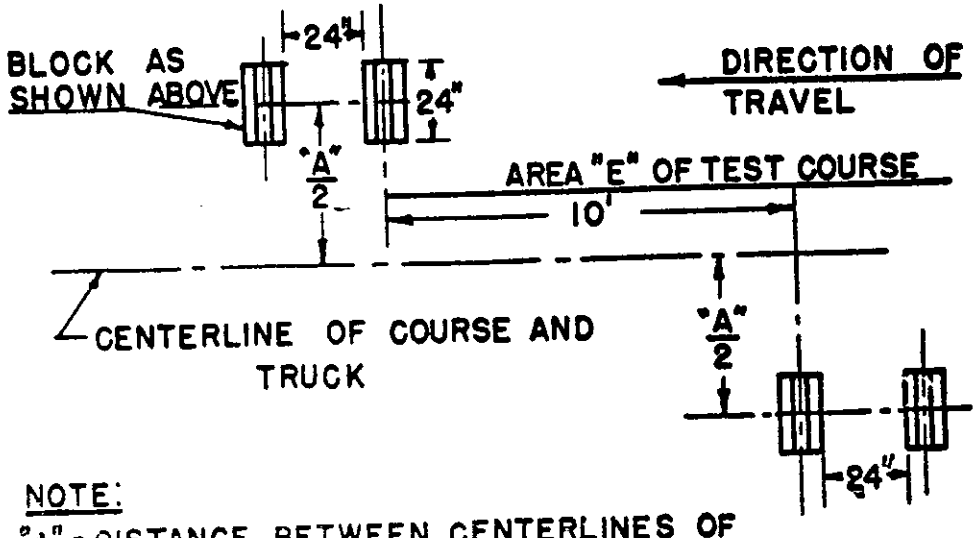
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DRAWING "B"

DETAILS OF OBSTACLE BLOCK CONSTRUCTION



LAYOUT OF OBSTACLE TEST

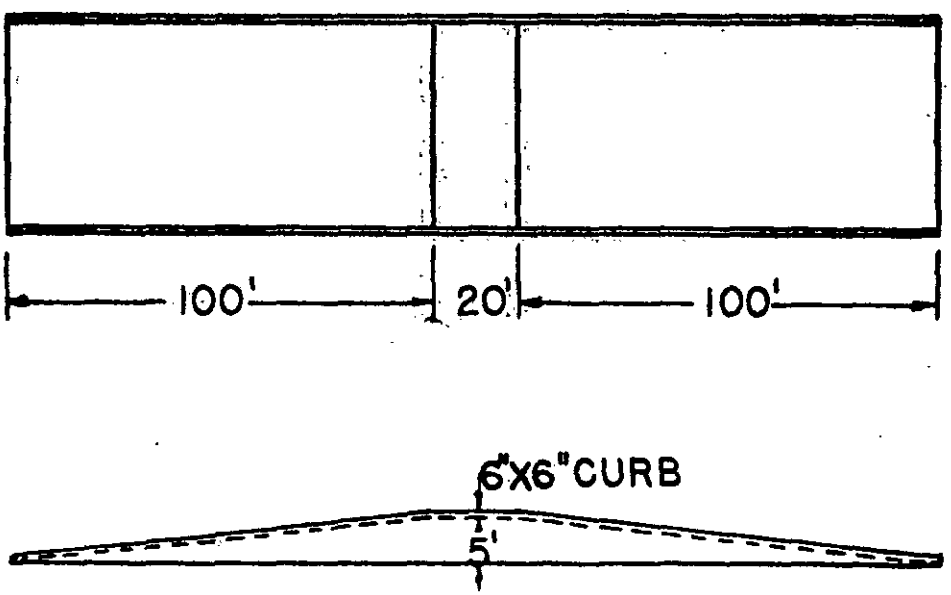


NOTE:
"A" = DISTANCE BETWEEN CENTERLINES OF DRIVING WHEELS

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DRAWING "C"

RAMP DESIGN



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Both the title and identifying symbol number should be stipulated when requesting copies of Military standards.

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